



India Lead Mass-Balance Model

Model Overview & Technical README

A parallel formal/informal accounting of lead flows through India's lead-acid battery economy, 2018–2023

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Internal working document

This is an early-stage model built on publicly available data and literature-derived assumptions. Several key parameters are first approximations / rough estimates intended to be replaced with field-collected data. The model is structured so that better inputs can be substituted without redesign; identifying which assumptions most require field validation is one of its central purposes.

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1 Purpose and scope

This document describes a country-level mass-balance model of lead flows through India’s lead-acid battery (LAB) economy, covering the years 2018–2023. The model’s purpose is **market-structure characterization**: understanding the scale of formal versus informal processing, identifying where unverified lead enters the chain, and testing whether domestic production reconciles with reported trade and refining statistics. It is *not* intended to produce precise tonnage estimates. Its outputs are best read as approximations and structural diagnostics rather than point figures.

The model traces lead around a closed loop. Batteries are installed into an in-service stock; the stock retires over time according to an effective battery lifetime; retired batteries are collected and processed back into refined lead through four sequential stages; and refined lead is manufactured into new batteries, which return to the stock. At each processing stage the model runs two parallel pathways — a **formal** (regulated, higher-recovery) lane and an **informal** (unregulated, lower-recovery) lane.

All quantities are expressed in tonnes of contained lead per year. Trade data are drawn from BACI bilateral statistics; refined and primary production from the US Geological Survey’s global mineral database; and the installed-stock series is a constructed bottom-up estimate from vehicle-fleet data. The model is calibrated over a smoothed window (a three-year centred rolling mean on the 2018–2023 series) to reduce year-to-year reporting noise.

1.1 Relationship to comparable methodologies

This model belongs to an established tradition of material-flow analysis (MFA) for lead, and draws on that literature for benchmarks and parameter priors. Dente et al. (2025) reconstruct in-use lead stocks across eleven countries over 1950–2018 and provide independent in-use-stock benchmarks. Dong et al. (2025) present a detailed China MFA for 2000–2023. Tür et al. (2016), through the Oeko-Institut, develop field-based methods for estimating the volumes of used lead-acid batteries generated and recycled in African settings, which inform the approach to the informal sector taken here. The present model is a country-level application in this tradition, combining bilateral trade data with exogenous production and independent stock estimates to characterize a national lead-recycling economy and estimate the possible scale of its informal sector.

2 The processing chain

2.1 Structure

Lead flows through four sequential stages — **breaking, smelting, refining, manufacturing** — each split into a formal and an informal lane. The loop is closed by retirement (the stock returning as collected scrap) and installation (manufactured batteries entering the stock). Figure 1 (final page) shows the full structure.

At each stage, a formal share φ governs the fraction of that stage’s domestically-sourced throughput routed through the formal lane, drawing from the combined output of both upstream lanes. Because the formal share rises down the chain, the formal lane at each stage draws more material than the previous formal stage produced; that excess is informal material sold into the formal

sector, captured as an *implied crossover* rather than a free parameter. This mirrors documented practice in which informal smelters sell crude lead to formal refiners.

DECISION

Formal share rises monotonically down the chain. We impose the ordering $\varphi_{\text{break}} < \varphi_{\text{smelt}} < \varphi_{\text{refine}} < \varphi_{\text{mfg}} < 1$. Informal processing is assumed to dominate at the dirty, low-capital end (breaking) and to diminish but never vanish toward the capital-intensive end (manufacturing). No stage is ever fully formal.

DECISION

External trade is formal-only. Imports and exports of used batteries, scrap, crude lead, feed materials, and finished batteries are assumed to move through legal customs channels and therefore enter the *formal* lane exclusively. The informal lane is purely domestic. Primary (mined) lead is likewise formal.

2.2 Equations

Retirement and collection. The stock growth rate g is fitted by log-linear regression on the installed-stock series. The retirement rate r follows the growth-corrected form, and collected lead is the collection rate γ applied to retirement:

$$r = \frac{g}{e^{g\tau} - 1}, \quad \text{RETIRE}(t) = k S(t) r, \quad C(t) = \gamma \text{RETIRE}(t). \quad (1)$$

Here $S(t)$ is the installed stock, τ the effective lifetime, and k a stock-scaling factor (default 1.0). Collection is a single shared pool; the formal/informal split occurs at breaking.

Breaking. With T^u the net (import minus export) used-battery trade:

$$b_F^{\text{in}} = \varphi_b C + T^u, \quad b_I^{\text{in}} = (1 - \varphi_b) C, \quad (2)$$

$$b_F^{\text{out}} = \delta \eta_{bF} b_F^{\text{in}}, \quad b_I^{\text{out}} = \delta \eta_{bI} b_I^{\text{in}}. \quad (3)$$

Smelting. The domestic scrap pool $P_s = b_F^{\text{out}} + b_I^{\text{out}}$ is split by φ_s ; net scrap trade T^s enters the formal lane. Outputs are crude lead:

$$s_F^{\text{in}} = \varphi_s P_s + T^s, \quad s_I^{\text{in}} = (1 - \varphi_s) P_s, \quad (4)$$

$$s_F^{\text{out}} = \eta_{sF} s_F^{\text{in}}, \quad s_I^{\text{out}} = \eta_{sI} s_I^{\text{in}}. \quad (5)$$

Refining. The crude pool $P_r = s_F^{\text{out}} + s_I^{\text{out}}$ is split by φ_r ; net crude trade T^c (HS 780199) enters the formal lane. The formal refined output R_F is the quantity anchored to USGS:

$$r_F^{\text{in}} = \varphi_r P_r + T^c, \quad r_I^{\text{in}} = (1 - \varphi_r) P_r, \quad (6)$$

$$R_F = \eta_{rF} r_F^{\text{in}}, \quad R_I = \eta_{rI} r_I^{\text{in}}. \quad (7)$$

Manufacturing. The refined pool $P_m = R_F + R_I$ is split by φ_m ; primary lead M^p and net feed trade T^f enter the formal lane. The battery share β applies to the refined-feed branch; battery parts G (HS 850790) are battery-committed and bypass β :

$$m_F^{\text{in}} = \varphi_m P_m + M^p + T^f, \quad m_I^{\text{in}} = (1 - \varphi_m) P_m, \quad (8)$$

$$\text{MFG}_F = \beta \eta_{mF} m_F^{\text{in}} + \eta_{mF} G_+, \quad \text{MFG}_I = \beta \eta_{mI} m_I^{\text{in}}. \quad (9)$$

Installation. Both lanes' output plus net finished-battery trade T^b give implied installation, compared against stock-derived demand:

$$\text{INSTALL}_{\text{impl}} = \text{MFG}_F + \text{MFG}_I + T^b, \quad \text{INSTALL}_{\text{tgt}} = \Delta S + \text{RETIRE}. \quad (10)$$

Implied crossovers. The informal-origin material drawn into the formal lane at each downstream boundary is

$$X_s = \varphi_s P_s - b_F^{\text{out}}, \quad X_r = \varphi_r P_r - s_F^{\text{out}}, \quad X_m = \varphi_m P_m - R_F. \quad (11)$$

A positive value is informal material sold up into the formal sector.

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Crossover sign conventions. A *negative* X_s or X_r means the formal lane cannot supply its own share even after the upstream split — a sign that the φ ordering is locally infeasible at that boundary, which the model flags. At manufacturing, X_m may be negative for a benign reason: large primary and feed inflows ($M^p + T^f$) can mean the formal lane does not need all formal-refined output. The manufacturing crossover is therefore reported as informational only.

2.3 Product categories and lead-content conversion factors

All trade quantities enter the model in tonnes of contained lead. Customs records report gross product mass, which must be converted to lead content using a product-specific factor; Table 1 gives the factor and basis for each HS code in the model. (BACI provides trade in lead-content terms directly.) In practice these factors may vary considerably, so further analysis will be needed to refine these estimates and account for their uncertainty.

The stage at which each product enters the chain is set by the equations of Section 2.2: scrap (780200) enters smelting, crude unwrought lead (780199) enters refining, refined lead and lead compounds enter the manufacturing feed pool, and finished batteries (850710 / 850720) bypass manufacturing and add directly to installation.

2.4 Treatment of specific trade codes

DECISION

HS code handling at manufacturing. Battery parts (HS 850790) are battery-committed and routed to formal manufacturing with the manufacturing efficiency but *outside* the battery-share β (they are already destined for batteries, so the share discount does not apply). Lead oxides (HS 282410 / 282490) remain in the feed pool and *are* subject to β (litharge could have non-battery uses). Finished batteries (HS 850710 / 850720) skip manufacturing entirely and add directly to installation.

3 Parameters

Table 2 lists the model's parameters, their current defaults, the basis for each, and whether each is held fixed, used as a diagnostic lever, or underdetermined by the data (see Section 4). Two pa-

HS	Product	Factor	Basis
260700	Ore / concentrates	0.60	Trade-weighted average
780110	Refined lead	1.00	99%+ purity
780191	Antimonial lead	0.95	1–6% Sb typical
780199	Other unwrought lead	0.95	Minor alloying
780200	Lead scrap	0.97	Metallic Pb scrap
850710	SLI batteries	0.65	34% Pb + 39% PbO paste
850720	Industrial batteries	0.70	Thicker plates
850790	Battery parts	0.80	Mostly plates / grids
854810	Waste batteries	0.70	Acid typically drained
282410	Lead oxides	0.91	Stoichiometric (PbO 92.8%)
282490	Other Pb compounds	0.75	Heterogeneous

Table 1: Lead-content conversion factors by HS product category. Factors convert gross traded mass to contained lead; all model flows are in lead-content tonnes.

parameters — the battery share β and the effective lifetime τ — carry most of the model's sensitivity and are the roughest first approximations.

The battery share $\beta = 0.86$ is the global-average battery share of refined-lead consumption. India-specific estimates have been lower; a lower β reduces manufacturing output and widens the installation gap, so this is a priority for field refinement. The effective lifetime $\tau_{\text{eff}} = 3.68$ years is computed by the same lifetime-weighted harmonic-mean method used in earlier versions of the model, but with shorter, field-derived per-segment life expectancies reflecting India's hot-climate operating conditions (a lead-acid battery's service life falls sharply with sustained high temperature). Both are defensible first approximations and both are explicit targets for further proposed field work.

DECISION

Effective lifetime from segment composition. τ_{eff} is the stock-weighted harmonic mean of per-segment lifetimes. The harmonic form is required because retirement scales with the reciprocal of lifetime; short-lived segments (e.g. e-rickshaw traction batteries) therefore dominate the retirement flow out of proportion to their share of installed stock.

4 What the model can and cannot tell you

The model is deliberately explicit about *identifiability* — which quantifies the data can actually pin down. It has more free parameters than independent anchors: four formal shares constrained by two anchors. As a consequence, the model identifies certain *combinations* of parameters sharply while leaving the individual values underdetermined. This is a property of the data, not a flaw in the method, and the model reports it rather than obscuring it.

Two structural features matter in particular. First, the stock scale and the effective lifetime enter the retirement calculation as a near-ratio: a range of (stock, lifetime) pairs produces nearly identical flows, so the data constrain their *ratio* well but the individual levels weakly. An external estimate

Parameter	Symbol	Default	Basis / status
Battery share of refined lead	β	0.86	Global-average share of refined lead consumed by the LAB sector; India-specific value a refinement target
Total collection rate	γ	0.98	Literature (total, formal + informal); held fixed
Stock-scaling factor	k	1.0	Identity; diagnostic lever
Effective lifetime	τ_{eff}	3.68 yr	Stock-weighted harmonic mean of segment lifetimes; field-derived
Formal shares	$\varphi_{b,s,r,m}$	0.70 / 0.80 / 0.90 / 0.95	Informed defaults, ordered; underdetermined (Section 4)
Formal efficiencies	η_F	0.95 / 0.97 / 0.99 / 0.98	Literature / engineering; held fixed
Informal efficiencies	η_I	0.70 / 0.60 / 0.95 / 0.95	Literature + field inference; informal manufacturing least certain
Breaking pre-factor	δ	0.95	Recovery at breaking; held fixed

Table 2: Model parameters and their current values. Efficiency rows list break / smelt / refine / mfg in order.

of either quantity — most tractably, lifetime from fleet composition — transfers precision to the other. Second, the four formal shares trade off against one another along the chain and are best reported as a feasible *family* consistent with the anchors and the ordering constraint, not as four independent point values.

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Practical consequence. The model's reliable outputs are structural and comparative — the scale of informal processing, the existence and size of an unrecorded refined-lead surplus, and the consistency or tension between reported stock and reported refining — rather than precise individual parameter values. Where a quantity is not separately identified, the model reports the constrained combination and the feasible range, not a single number.

5 Known limitations and open items

This is a proof-of-concept model, and its limitations define the work it is intended to launch.

The stock series is not yet well validated against field data. It is constructed bottom-up from vehicle-fleet data and carries known upward biases: registered-versus-active fleet counts (registrations include vehicles no longer on the road), a stock-versus-flow approximation in the segment gross-up, and an unpinned motorcycle-battery lead mass that has high leverage given the size of the two-wheeler fleet. A more accurate installed-lead estimate is the single highest-value input the proposed work would produce.

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Anchor reconciliation. The installation and refining anchors are sensitive to the stock series and to β and τ . At the current defaults they sit within roughly 8% of joint balance, but that proximity depends on rough inputs; under alternative plausible values the two anchors diverge, implying that at least one reported input — installed stock, the battery share, or reported secondary production — requires correction. Quantifying and reconciling this tension with field data is a central objective of the proposed work, and the model's structure is designed to localize where the correction must fall.

Informal manufacturing efficiency is the least certain efficiency assumption. It is currently set close to the formal value; if informal assembly is materially less efficient, manufacturing output falls and the installation gap widens. By contrast, informal *refining* efficiency does not affect the USGS-anchored solve — the formal refined output is independent of it — which the model surfaces explicitly. This is a useful finding in itself: informal refining efficiency is not a lever for the refining anchor and need not be a field-research priority.

Formality is modelled stage-by-stage. Field evidence suggests the boundary may instead be operator-structured: operators tend to be formal or informal across all stages, with some selling of informal output into formal channels. The implied-crossover mechanism approximates this, but a more faithful operator-level structure is a candidate refinement once field data on operator behaviour are available.

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Trade attachment. The precise point at which formal-only trade enters each stage (before or after the domestic formal/informal split) is a modelling convention rather than an observed quantity. The current treatment adds net trade to the formal lane after the split; field data on the formality of importers and exporters would let this be set empirically.

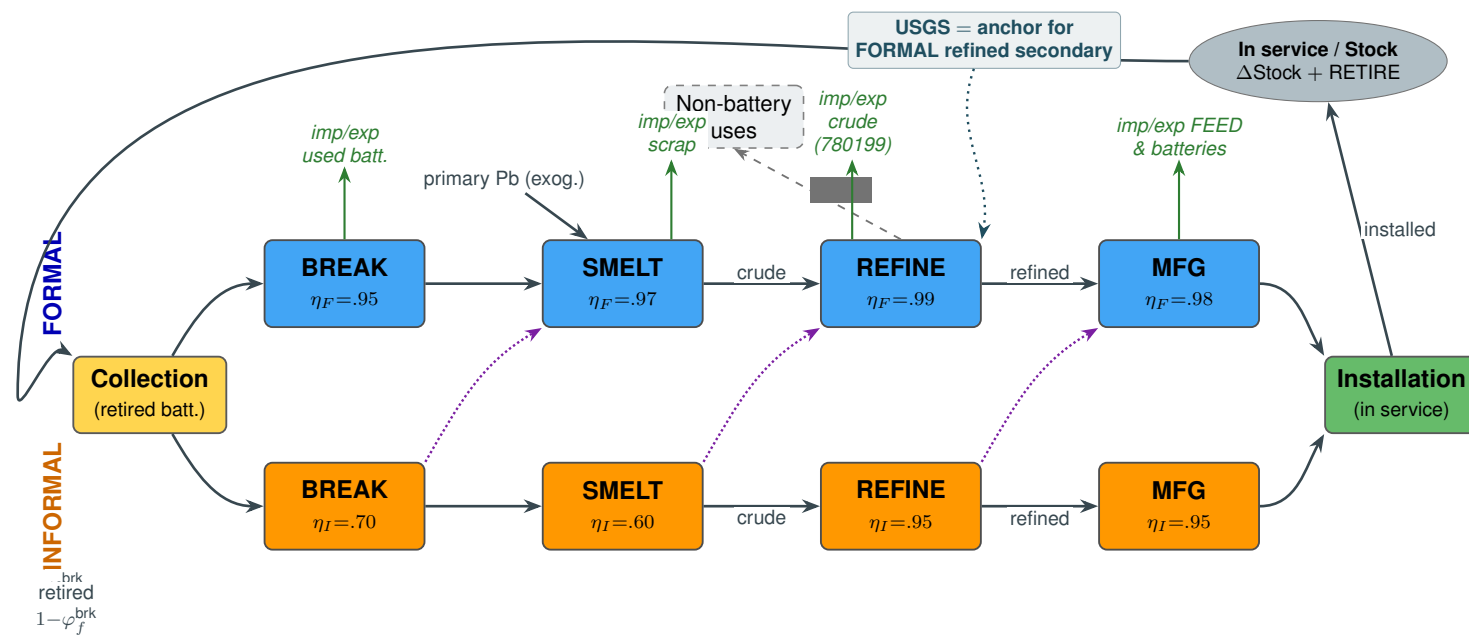
Vehicle imports and exports are not yet accounted for. The model currently treats the installed-stock series as the basis for retirement and does not separately track lead entering or leaving the country embodied in whole vehicles (new or used vehicle trade). We judge that this flow is unlikely to shift the balance of the model materially at present, and have intentionally deferred it; it will be incorporated in the next layer of refinement.

6 References

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Figure 1. Parallel formal/informal lead-flow structure



Formal share rises down the chain: $\varphi_f^{brk} < \varphi_f^{smelt} < \varphi_f^{refine} < \varphi_f^{mfg} < 1$

Legend: $\longrightarrow$ domestic flow $\cdots \longrightarrow$ informal $\rightarrow$ formal (sold up) $\longrightarrow$ trade $\longrightarrow$ formal chain only $\cdots \longrightarrow$ anchor (not a flow)